THE STRUCTURE MORPHISM S_U -> [U/R] IS SMOOTH AND SURJECTIVE FOR A SMOOTH GROUPOID

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the final identification is 'Clearly' from the established descriptions (mechanical recognition). The diagonal input lemma is not carded (two-line citation) and is declared external.

Allowed standard prerequisites (editorial).

- Representability of the fibre product. lemma-diagonal-quotient-stack (same chapter, NOT carded, declared external) and lemma-representable-diagonal (carded L26680). S_U x_{[U/R]} (Sch/T) is representable by an algebraic space F.
- 2-fibre product algebra. Categories, lemma 2-fibre-product-erase-factor (cross-chapter, named). First rewriting equality.
- Quotient stack 2-cartesian square. Groupoids in Spaces, lemma quotient-stack-2-cartesian (cross-chapter, named). Second rewriting equality: S_U x_{[U/R]} S_U = S_R.
- Smoothness of s with a section. the groupoid hypothesis s: R -> U smooth with identity section e: U -> R (structural). R x_{s,U,x} T -> T is smooth and surjective.

Source reference. Original reference 02XD: focused-seeds/geometry/S9-alg-smooth-quotient-smooth-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04WZ: sources/stacks-algebraic/pinned-original-source.tex, source lines 2571–2571. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045G: sources/stacks-algebraic/pinned-original-source.tex, source lines 1337–1337. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04M9: focused-seeds/geometry/S9-alg-smooth-quotient-smooth-presentation/ref-tags.txt, source lines 6–6. Navigation pointer

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let (U, R, s, t, c) be a smooth groupoid in algebraic spaces over S. Then the morphism $\mathcal{S}_U \to [U/R]$ is smooth and surjective.*

Proof. Let T be a scheme and let $x : (Sch/T)_{fppf} \to [U/R]$ be a 1-morphism. We have to show that the projection

$$\mathcal{S}_U \times_{[U/R]} (Sch/T)_{fppf} \longrightarrow (Sch/T)_{fppf}$$

is surjective and smooth. We already know that the left hand side is representable by an algebraic space F, see Lemmas 04WZ and 045G. Hence we have to show the corresponding morphism $F \to T$ of algebraic spaces is surjective and smooth. Since we are working with properties of morphisms of algebraic spaces which are local on the target in the fppf topology we may check this fppf locally on T. By construction, there exists an fppf covering $\{T_i \to T\}$ of T such that $x|_{(Sch/T_i)_{fppf}}$ comes from a morphism $x_i : T_i \to U$. (Note that $F \times_T T_i$ represents the 2-fibre product $\mathcal{S}_U \times_{[U/R]} (Sch/T_i)_{fppf}$ so everything is compatible with the base change via $T_i \to T$.) Hence we may assume that x comes from $x : T \to U$. In this case we see that

$$\mathcal{S}_U \times_{[U/R]} (Sch/T)_{fppf} = (\mathcal{S}_U \times_{[U/R]} \mathcal{S}_U) \times_{\mathcal{S}_U} (Sch/T)_{fppf} = \mathcal{S}_R \times_{\mathcal{S}_U} (Sch/T)_{fppf}$$

The first equality by Categories, Lemma 02XD and the second equality by Groupoids in Spaces, Lemma 04M9. Clearly the last 2-fibre product is represented by the algebraic space $F = R \times_{s,U,x} T$ and the projection $R \times_{s,U,x} T \to T$ is smooth as the base change of the smooth morphism of algebraic spaces $s : R \to U$. It is also surjective as s has a section (namely the identity $e : U \to R$ of the groupoid). This proves the lemma. □